End-to-End MLOps Pipeline Architecture

Project Overview

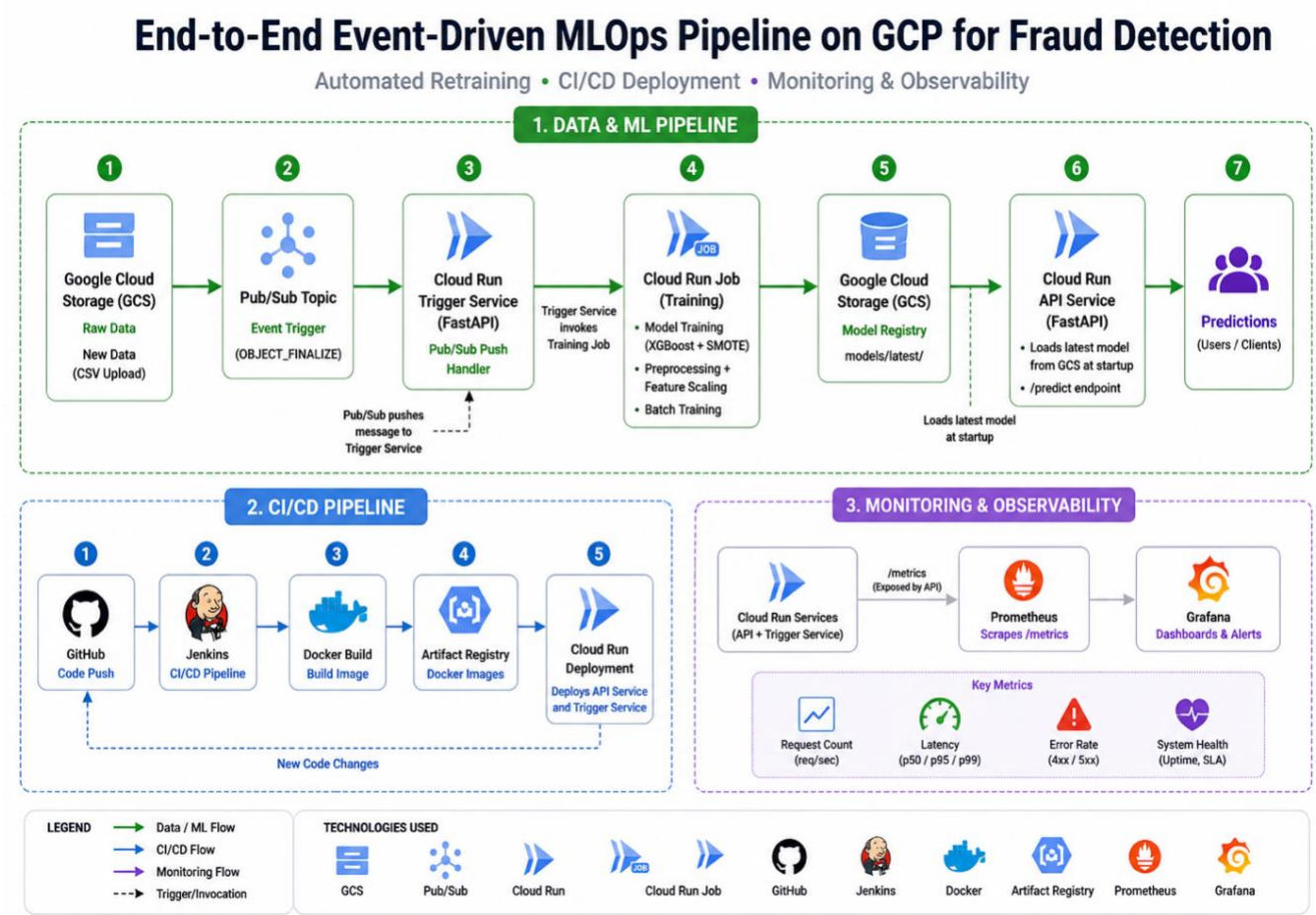
This project implements a fully event-driven, serverless MLOps pipeline on Google Cloud Platform for credit card fraud detection.

The system automatically retrains a machine learning model when new data arrives, deploys updated services via CI/CD, and provides real-time monitoring of API performance.

Key capabilities include:

- Event-driven retraining using GCS + Pub/Sub + Cloud Run Jobs
- Automated CI/CD pipeline using Jenkins, Docker, and Artifact Registry
- Scalable API deployment on Cloud Run with dynamic model loading
- Real-time observability using Prometheus and Grafana

System Architecture



Workflow Explanation

1. Data Pipeline (Event-Driven Retraining)

- New transaction data is uploaded to Google Cloud Storage (GCS)
- A Pub/Sub OBJECT_FINALIZE event is triggered
- A Cloud Run Trigger Service (FastAPI) receives the event
- The trigger service programmatically invokes a Cloud Run Job
- The job performs:
- Data preprocessing
- Feature scaling
- Model training (XGBoost + SMOTE)
- The trained model is stored in GCS under models/latest/

2. CI/CD Pipeline (Automated Deployment)

- Code changes are pushed to GitHub
- Jenkins pipeline is triggered manually or via webhook
- Pipeline steps:
- Build Docker images for API and trigger services
- Push images to Google Artifact Registry
- Deploy updated services to Cloud Run
- Ensures consistent, repeatable deployments without manual intervention

3. Monitoring Pipeline & Observability

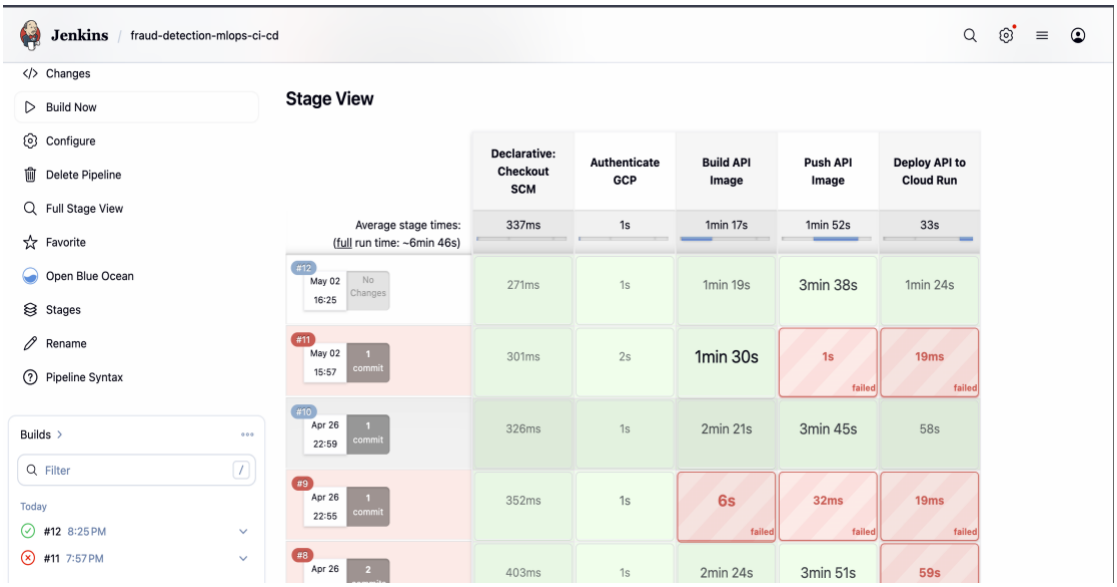
- The FastAPI service exposes a /metrics endpoint
- Prometheus scrapes application metrics at regular intervals
- Grafana visualizes system performance through dashboards
- Tracked metrics include:
    - API request rate (requests/sec)
    - Prediction traffic (/predict endpoint)
    - Error rate (4xx responses)
    - Prediction success rate (% of successful responses)

4. System Behavior

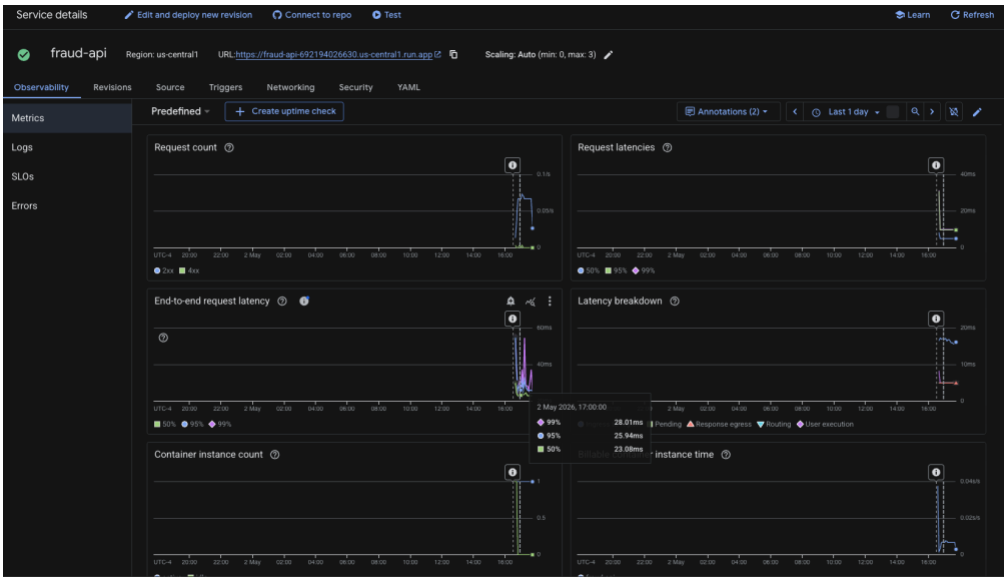
- The API dynamically loads the latest model from GCS at startup
- New data automatically triggers retraining without human intervention
- Updated models are immediately available for inference after training
- The system scales automatically based on incoming request traffic (Cloud Run autoscaling)

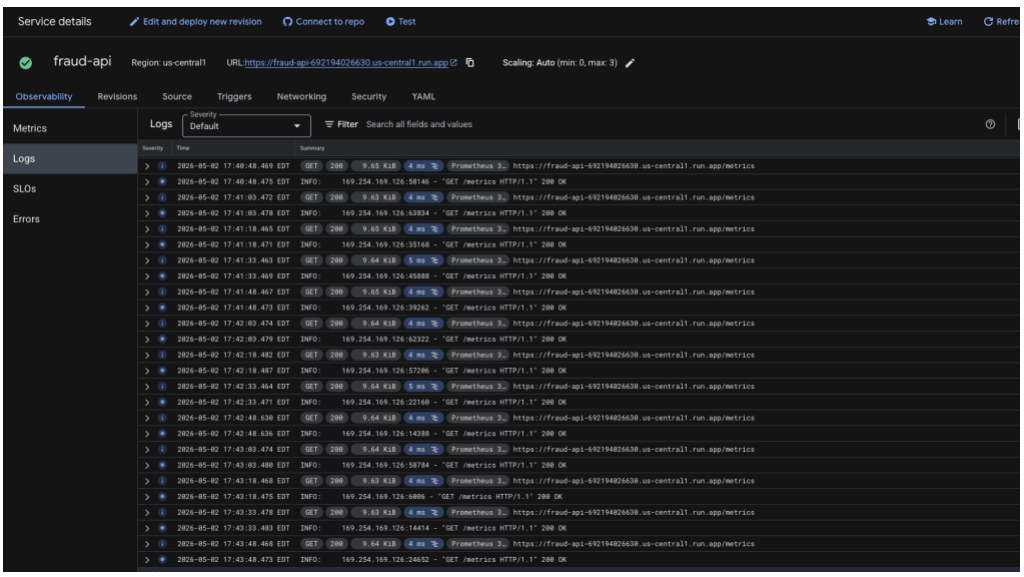
Screenshots

Jenkins CI/CD Pipeline (Shows automated build, push, and deployment stages)

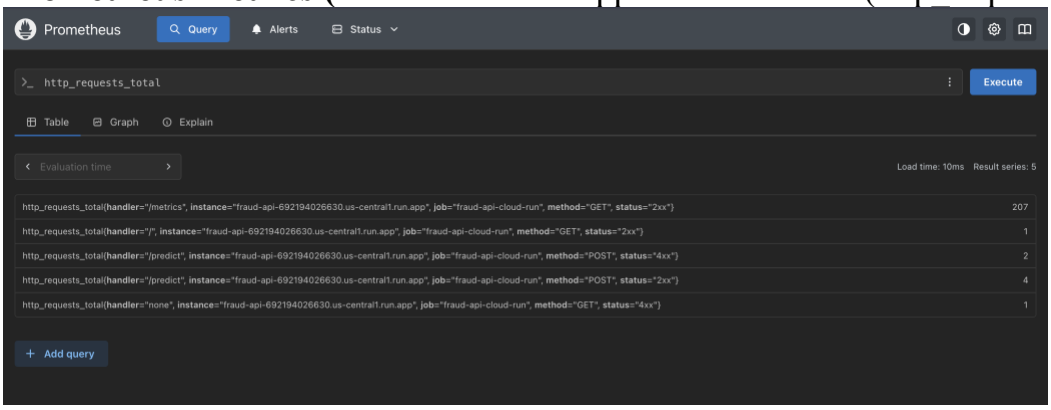


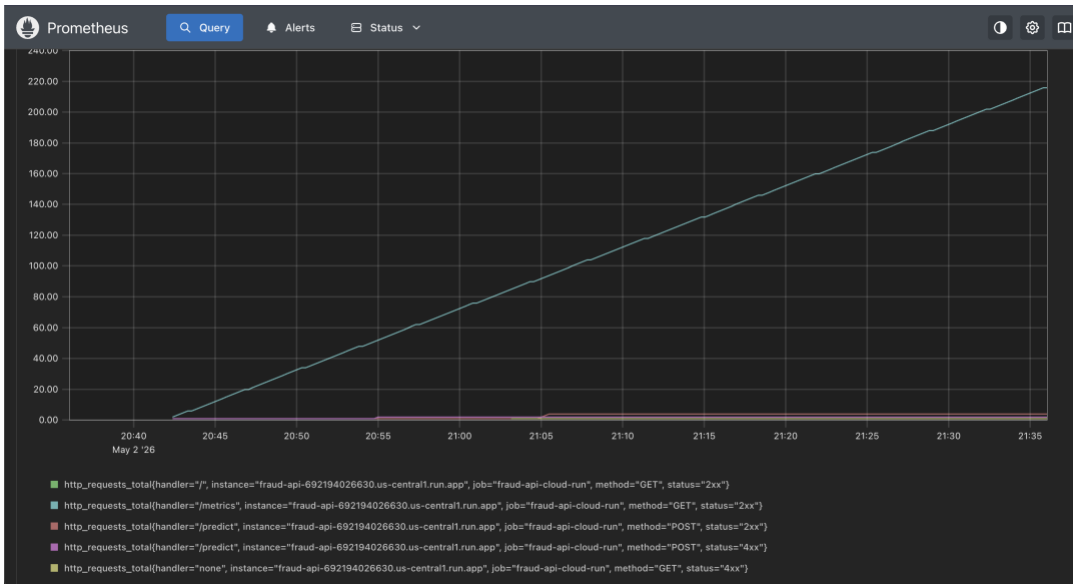
Cloud Run Deployment(Shows API service, trigger service, and training job)





Prometheus Metrics (Shows collected application metrics (http_requests_total**)**





Grafana Dashboard (Shows real-time monitoring of API performance and system health)

Key Highlights

- Fully automated **event-driven retraining pipeline**
- CI/CD with **Jenkins + Docker + Cloud Run**
- Monitoring using **Prometheus + Grafana**
- Production-style architecture using **GCP services**

Tech Stack
GCP: Cloud Run, GCS, Pub/Sub, Artifact Registry
ML: XGBoost, SMOTE, Scikit-learn
Backend: FastAPI
CI/CD: Jenkins, Docker
Monitoring: Prometheus, Grafana

This architecture follows production-grade MLOps patterns used in real-world systems, including event-driven pipelines, containerized deployments, and observability-first design.

Author

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